FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no.22

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Recap *of previous lecture*

P & C



Topics to be covered

"PERCENTAGE"



(*) a is how much % of b

$\Rightarrow a = \frac{?}{100} \times b$

$\boxed{? = \frac{100a}{b}}$

eg 40 is how much % of 200

$\Rightarrow 40 = \frac{?}{100} \times 200$

$? = \frac{40 \times 100}{200} = 20\%$

eg 35 is how much % of 140.

$\Rightarrow 35 = \frac{?}{100} \times 140$

$\text{ie } ? = \frac{35 \times 100}{140} = \frac{100}{4} = 25\%$

eg: 27 is how much % 81 ?

$\Rightarrow 27 = \frac{?}{100} \times 81 \quad \text{ie } ? = \frac{27 \times 100}{81} = \frac{100}{3}$

$= 33.33\%$

Eg Ram's Salary is 90% of Shyam's Salary then Shyam's Salary is how much % of Ram?

(M-I) $R = 90\% \text{ of } S$

$R = \frac{90}{100} \times S$

$S = \frac{100}{90} R$ — (1)

Now S is how much % of R

$\Rightarrow S = \frac{?}{100} \times R$

$? = \frac{100}{R}(S) = \frac{100}{R}\left(\frac{100}{90}R\right) = 111.11\%$

(M-II) ATQ, $R = \left(\frac{90}{100}\right)S$

$\Rightarrow S = \left(\frac{100}{90}\right)R = \left(\frac{10}{9}\right)R = \left(1 + \frac{1}{9}\right)R$

i.e. $S = \left(1 + \frac{1}{9}\right)R$

$\frac{1}{9} = 11.11\%$

So S Salary is 111.11% of R. Salary

% Fraction Equivalence

$1 = 100\%$	$\frac{1}{6} = 16.66\%$	$\frac{1}{11} = 9.09\%$	$\frac{1}{20} = 5\%$
$\frac{1}{2} = 50\%$	$\frac{1}{7} = 14.28\%$	$\frac{1}{12} = 8.33\%$	$\frac{1}{25} = 4\%$
$\frac{1}{3} = 33.33\%$	$\frac{1}{8} = 12.5\%$	$\frac{1}{14} = 7.14\%$	
$\frac{1}{4} = 25\%$	$\frac{1}{9} = 11.11\%$	$\frac{1}{15} = 6.66\%$	
$\frac{1}{5} = 20\%$	$\frac{1}{10} = 10\%$	$\frac{1}{16} = 6.25\%$	

% Increase → if x is Increased by 20; then (M.f) $= 1 + \frac{1}{5} = \frac{6}{5}$ & F.Value $= \frac{6}{5}x$

% Decrease → if x is decreased by 6.66% then (Mf) $= 1 - \frac{1}{15} = \frac{14}{15}$ & Final Value $= \frac{14}{15}x$

eg 37.5% of 240 = ?

$\Rightarrow \frac{3}{8} \times 240 = 90$

eg 22.22% of 540 = ?

$\Rightarrow \frac{2}{9} \times 540 = 120$

eg 14.28% of 350 = ?

$\Rightarrow \frac{1}{7} \times 350 = 50$

eg: If 240 is Increased by 37.5% then F.Value = ?

Sol: $37.5\% = \frac{3}{8}$ so $MF = 1 + \frac{3}{8} = \frac{11}{8}$

$\therefore$ F.Value $= \frac{11}{8} \times 240 = 330$

eg If 540 is decreased by 22.22% then F.Value = ?

Sol: $22.22\% = \frac{2}{9}$ so $MF = 1 - \frac{2}{9} = \frac{7}{9}$

$\therefore$ F.Value $= \frac{7}{9} \times 540 = 420$

eg If 350 is Increased by 14.28% then F.Value = ? $= \left(1 + \frac{1}{7}\right) 350 =$

eg " " decreased by 14.28% then F.Value $= \left(1 - \frac{1}{7}\right) 350 =$

Q: if x is 20% more than y then y is what % less than x?

Sol: $20\% \simeq \pm\frac{1}{5}$ so ATQ

$$x = \left(1+\frac{1}{5}\right)y = \frac{6}{5}y$$

$$y = \frac{5}{6}x$$

$$y = \left(1-\frac{1}{6}\right)x \quad \frac{1}{6} = 16.66\%$$

so y is 16.66% less than x

Q: 120 is 20% more than 100
so 100 is 16.66% % less than 120

$$100 = \frac{?}{100} \times 120 \Rightarrow ? = \frac{100\times100}{120} = 83.33\%$$

Q: If the height of a Δ is decreased by 10%, it area still increases by 10% then find % change in it's Base?

$10\% = \frac{1}{10}$

$f_I = 1+\frac{1}{10} = \frac{11}{10}$

$f_D = 1-\frac{1}{10} = \frac{9}{10}$

$A' = \frac{11}{10}A$

$H' = \frac{9}{10}H$

$B' = ?$

$\because A = \frac{1}{2}B\times H$ — (1)

so $A' = \frac{1}{2}B'\times H'$

$\frac{11}{10}A = \frac{1}{2}B'\left(\frac{9}{10}H\right)$ — (2)

$\frac{11}{10}\left(\frac{1}{2}BH\right) = \frac{1}{2}B'\left(\frac{9}{10}H\right)$

Sol: $10\% = \frac{1}{10}$ → $f_I = 1 + \frac{1}{10} = \frac{11}{10}$, $f_D = 1 - \frac{1}{10} = \frac{9}{10}$

$A = \frac{1}{2} B \times H$ ——— (1)

$A' = \frac{11}{10} A$ & $H' = \frac{9}{10} H$

$A' = \frac{1}{2} B' \times H'$

$\left(\frac{11}{10} A\right) = \frac{1}{2} B' \left(\frac{9}{10} H\right)$

$\frac{11}{10}\left(\frac{1}{2} BH\right) = \frac{1}{2} B' \left(\frac{9}{10} H\right)$

$B' = \frac{11}{9} B = \left(1 + \frac{2}{9}\right) B$ → 22.22%

Q: The population of a city is increased by 25%, 18.18%, 10% in 3 consecutive censuses but decreased by 9.09% in 4th census & it is now 65 lacs then find the pop before 1st census?

$+25\% = +\frac{1}{4} \Rightarrow f_1 = 1 + \frac{1}{4} = \frac{5}{4}$

$+18.18\% = +\frac{2}{11} \Rightarrow f_2 = 1 + \frac{2}{11} = \frac{13}{11}$

$+10\% = +\frac{1}{10} \Rightarrow f_3 = 1 + \frac{1}{10} = \frac{11}{10}$

$-9.09\% = -\frac{1}{11} \Rightarrow f_4 = 1 - \frac{1}{11} = \frac{10}{11}$

Let the population before 1st Census was x

A.T.Q.

$$\left[\left\{(x \times f_1) \times f_2\right\} \times f_3\right] \times f_4 = 65\,L$$

$$\left[\left\{\left(x \times \frac{5}{4}\right) \times \frac{13}{11}\right\} \times \frac{11}{10}\right] \times \frac{10}{11} = 65\,L$$

$$x = \frac{65\,L \times 11 \times 4}{5 \times 13} = 44\ \text{Lacs}$$

Method-II Cross check

→ HW

Q: If Rs 100 is decreased by (10%) successively then Final value = ?

Sol: (M-I) 100 Rs $\xrightarrow{-10\%}$ 90 Rs $\xrightarrow{-10\%}$ 81 Rs.

(M-II) $10\% = \frac{1}{10} \Rightarrow f = 1 - \frac{1}{10} = \frac{9}{10}$ So f. Value $= \left[(100) \times \frac{9}{10}\right] \times \frac{9}{10} = 81$ Rs.

(M-III) Single % change $= \left[-10 - 10 + \frac{(-10)(-10)}{100}\right]\% = -19\%$

$\therefore$ original Value = 100 Rs So f. Value = 81 Rs.

Note: Single % equivalence to two successive % changes of a % & b % can be given as;

$$\text{Single \% Change} = \left(a + b + \frac{ab}{100}\right)\%$$

Eg A shopkeeper increases S.P by 10% & then decreases by 10% then what is this % P or Loss ?

Sol: (M-I) Rs 100 $\xrightarrow{+10\%}$ 110 Rs $\xrightarrow{-10\%}$ 99 Rs is 1% Loss

(M-II) $10\% = \frac{1}{10}$ $\begin{cases} f_I = \frac{11}{10} \\ f_D = \frac{9}{10} \end{cases}$ $\Rightarrow$ New SP $= \left[(x) \times \frac{11}{10}\right] \times \frac{9}{10} = \frac{99}{100}x = \left(1 - \frac{1}{100}\right)x$ = 1% Loss

(M-III) Single % Change $= \left[+10 - 10 + \frac{(10)(-10)}{100}\right]\% = -1\%$ is 1% Loss

Q- A shopkeeper 1st decreases goods by 20% & then increases by 10%, Find his P&L %

(M-I) 100 $\xrightarrow{-20\%}$ 80 Rs $\xrightarrow{+10}$ 88 Rs i.e 12% loss

(M-II) $-20\% = -\frac{1}{5} \Rightarrow f_D = 1 - \frac{1}{5} = \frac{4}{5}$ & $+10\% = +\frac{1}{10} \Rightarrow f_I = 1 + \frac{1}{10} = \frac{11}{10}$

Hence f-Value $= \left(x \times \frac{4}{5}\right) \times \frac{11}{10} = \frac{44}{50}x = \frac{88}{100}x = \left(1 - \frac{12}{100}\right)x$

12% loss

(M-III) Single % change $= \left[-20 + 10 + \frac{(-20)(10)}{100}\right]\% = -12\%$

i.e 12% loss

Q. In an election 10% of the voters did not cast their vote, & 10% votes were Invalid. If winner got 70% of the Valid votes and won by 7290 votes then find the total number of voters in voting list?

Let Total No. of voters are $= x$

ATQ, $\left[\underbrace{\left(x \times \frac{9}{10}\right) \times \frac{9}{10}}_{\text{Valid Votes}}\right] \times \frac{70}{100} - \left[\underbrace{\left(x \times \frac{9}{10}\right) \times \frac{9}{10}}_{\text{Valid Votes}}\right] \times \frac{30}{100} = 7290$

$$\frac{81x}{100} \times \frac{(40)}{100} = 7290 \Rightarrow x = 22500$$

100% Voters $\xrightarrow{-10\%}$ 90% Votes $\xrightarrow{-10\%}$ 81% Valid Votes

% Diff of winner & loosers = 70% − 30% = 40% of Valid Votes

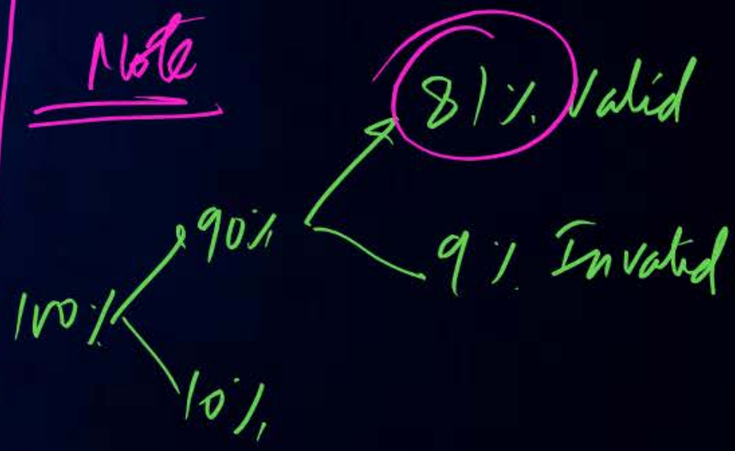
So Ans: 40% of 81% of Total Voters = 7290

$$\frac{40}{100} \times \frac{81}{100} \times \text{T.Voters} = 7290$$

$$\text{T.Voters} = \frac{7290 \times 100 \times 100}{40 \times 81}$$

$$= 22500 \text{ Voters}$$

Note



Q: When prices of Milk went up by 18.18%, household decreased it's consumption by 8.33% then find the % change in the expenditure of family?

$+18.18 = \frac{+2}{11} \Rightarrow f_p = 1 + \frac{2}{11} = \frac{13}{11}$

$-8.33 = \frac{-1}{12} \Rightarrow f_q = 1 - \frac{1}{12} = \frac{11}{12}$

$P' = \frac{13}{11}P$ & $Q' = \frac{11}{12}Q$

$E = PQ$ ——— (1)

$E' = P'Q'$

$E' = \left(\frac{13}{11}P\right)\left(\frac{11}{12}Q\right) = \frac{13}{12}(PQ)$

$E' = \left(1 + \frac{1}{12}\right)E$ (By (1))

$\frac{1}{12} = 8.33$

i.e Exp of the family Increases by 8.33%


Thank
YOU